Algebra II with Trigonometry Chapter 7.3

Mental Math (Gemini)

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Double-Angle Sine Evaluation 1. What is the exact numerical value of $2\sin(15^\circ)\cos(15^\circ)$?

Double-Angle Cosine Evaluation 2. What is the exact numerical value of $\cos^2(22.5^\circ) - \sin^2(22.5^\circ)$?

Double-Angle Tangent Evaluation 3. What is the exact numerical value of $\frac{2\tan(22.5^\circ)}{1-\tan^2(22.5^\circ)}$?

Lowering Powers Evaluation 4. Using the alternative forms of the cosine double-angle formula, what is the exact numerical value of $1 - 2\sin^2(15^\circ)$?

Simplifying Double-Angles 5. If $\sin(2x) = 2\sin(x)\cos(x)$, what is the simplified, single-term form of $2\sin(4\theta)\cos(4\theta)$?

Half-Angle Formula Recognition 6. Assuming the result is positive, what single trigonometric function is equivalent to $\sqrt{\frac{1-\cos(6x)}{2}}$?

Recognizing Power-Reducing Formulas 7. Rewrite the fraction $\frac{1+\cos(2x)}{2}$ as a single, squared trigonometric function of x.

Basic Identity Simplification 8. Simplify the fraction $\frac{\sin(2x)}{\cos(x)}$ into a single term.

Sum-to-Product Conversion 9. Using the sum-to-product formulas, how do you express $\sin(3x) + \sin(x)$ as a product of two trigonometric functions?

Conceptual Double-Angle Substitution 10. If you are given that $\cos(x) = 0$, what is the value of $\cos(2x)$? (Hint: Use a double-angle formula to solve this without calculating the angle x).